

WASP-135b

R-0184

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250705 · created 2026-07-16T19:41:42+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460861.706 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.05
Transit depth (Rp/Rs)^2	2.9 +/- 1.71 [%]
Orbital Inclination (inc)	86.82 +/- 2.9
Airmass coefficient 1 (a1)	0.56 +/- 0.22
Airmass coefficient 2 (a2)	0.47 +/- 0.85
Scatter in the residuals of the lightcurve fit is	37.67 %
Best Comparison Star	None
Optimal Aperture	2.06
Optimal Annulus	7.54
Transit Duration (day)	0.081 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.05 vs 0.139 (deviation 0.61σ)
Duration fit vs published	0.081 d vs 0.0687 d (deviation 1.09σ)
Mid-time O–C	-35.1 min
Depth SNR	3.3
Residual scatter	37.67 %

Light curve

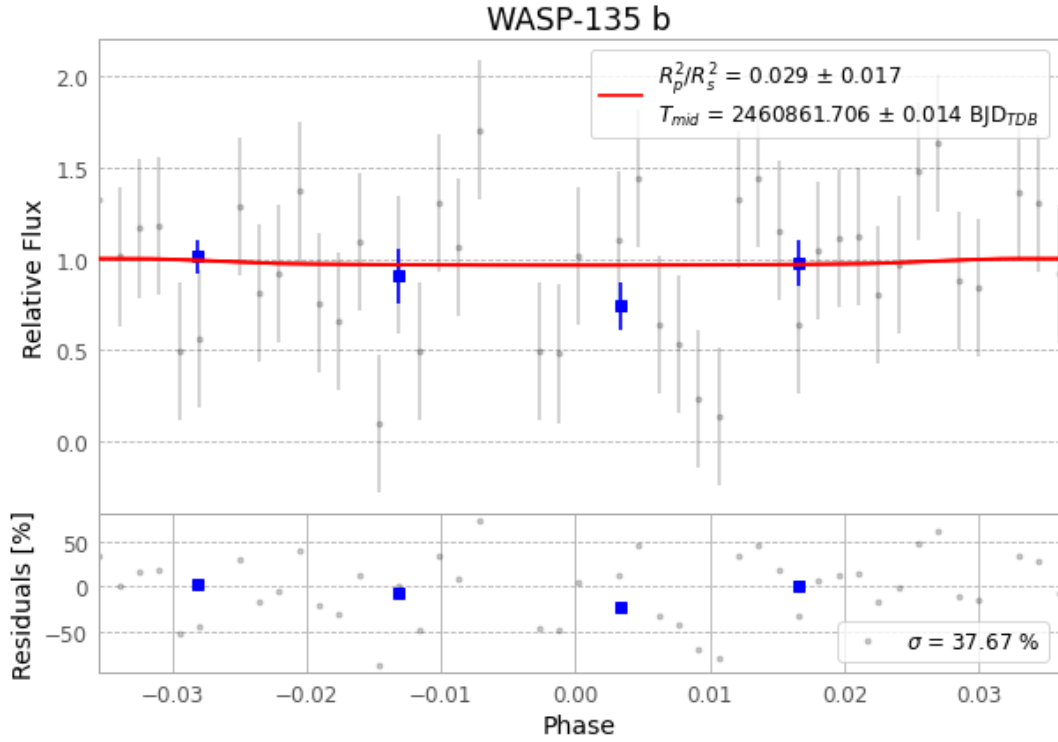


Figure 1 — FinalLightCurve_WASP-135 b_2025-07-04.png (SHA-256 d54264786d0cb814...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [514, 112], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fb78e5f34915a92a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

49 FITS frames (32 MB), WASP-135250705033915.FITS → WASP-135250705060315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250705.log (9f4889ac7c16...), FinalLightCurve_WASP-135 b_2025-07-04.png (d54264786d0c...), FinalLightCurve_WASP-135 b_2025-07-04.csv (a6517bdf958...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:41Z.